

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Defined and validated power architectures including utility intake, MV/LV distribution, and UPS topologies for a major data center campus expansion.
- Led power systems studies, including load flow and short circuit analysis, to ensure system stability and compliance with IEC/BS/EN standards.
- Integrated monitoring and controls systems (BMS/EPMS) with data centre infrastructure, aligning with Tier III availability objectives.
- Assessed and embedded decarbonization strategies, such as high-efficiency UPS and grid-interactive controls, for new data centre builds.
- Coordinated multi-disciplinary design interfaces, including mechanical, structural, and ICT, to deliver fully integrated data centre solutions.
- Mentored a team of 4 junior engineers on electrical design principles and best practices within a fast-track project environment.

SKILLS

Electrical Design – Power Architectures, MV/LV Distribution, UPS Topologies, Generators, Switchgear, PDUs, STS, Busway, Earthing, Lighting

Power Systems Analysis – Load Flow, Short Circuit, Protection Coordination, Arc-Flash, Harmonic Analysis, ETAP, SKM

Data Centre Technologies – Hyperscale, Mission Critical Environments, Tier II-IV Availability, Resilience Features, Decarbonization, Energy Efficiency

Standards & Compliance – IEC, BS, EN, NEC, NFPA, Client Specifications, AECOM Quality Procedures

Project Management – Technical Delivery, Program Management, Work Winning, Team Leadership, Financial Management, Commercial Acumen

Cross-Disciplinary Coordination – Mechanical, Structural, Architectural, Fire/Life Safety, Security, ICT, BMS, EPMS, SCADA

PROJECTS

Data Centre Electrical Resilience Upgrade (HCL Technologies)

- Engineered a comprehensive upgrade plan for electrical distribution and UPS systems across a multi-site data centre portfolio, improving resilience and reducing single points of failure.
- Performed detailed site surveys and conducted power quality analysis using specialized equipment to identify critical areas for intervention, leading to a 30% reduction in reported power anomalies.
- Collaborated with client operations teams to define new monitoring requirements and integrate enhanced alarm management into the existing EPMS, cutting diagnostic time by approximately 20%.

Energy Efficiency Assessment for Enterprise Data Hall (HCL Technologies)

- Analyzed existing power usage patterns and identified opportunities for energy efficiency improvements in a large enterprise data hall, focusing on cooling and power distribution.
- Modeled the impact of implementing advanced control strategies and higher efficiency UPS systems, projecting a potential energy saving of 15% on power infrastructure costs.
- Developed a technical report and presentation for client stakeholders, outlining recommended upgrades and their expected ROI, which informed their sustainability roadmap.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)